

Probiotics as prevention and treatment for diarrhea.

PURPOSE OF REVIEW: To critically appraise evidence on probiotic use for prevention and treatment of diarrhea in children and adults. **RECENT FINDINGS:** Several randomized controlled trials and meta-analyses suggested that probiotics are effective in primary and secondary prevention of gastroenteritis and its treatment. Selected *Lactobacillus* strains had a modest, although significant effect in primary prevention. *Saccharomyces boulardii* was effective in antibiotic-associated and in *Clostridium difficile* diarrhea. There is evidence that it might prevent diarrhea in day-care centers. *Lactobacillus rhamnosus* GG was associated with reduced diarrheal duration and severity, more evident in case of childhood Rotavirus diarrhea. Similar, although weaker, evidence was obtained with *S. boulardii*. Both strains are included in evidence-based recommendations for gastroenteritis management in children. Data on other *Lactobacillus* strains are preliminary. Probiotic efficacy was related to cause, early administration and bacterial load, and their mechanisms were associated with antiinfectious action in the intestine or, indirectly, to modulation of innate and adaptive immunity. **SUMMARY:** Probiotics have gained a role as adjunctive treatment of infantile gastroenteritis together with rehydration. Their efficacy is less convincing in adults, but promising in antibiotic-associated diarrhea. However, evidence of efficacy is limited to a few strains.